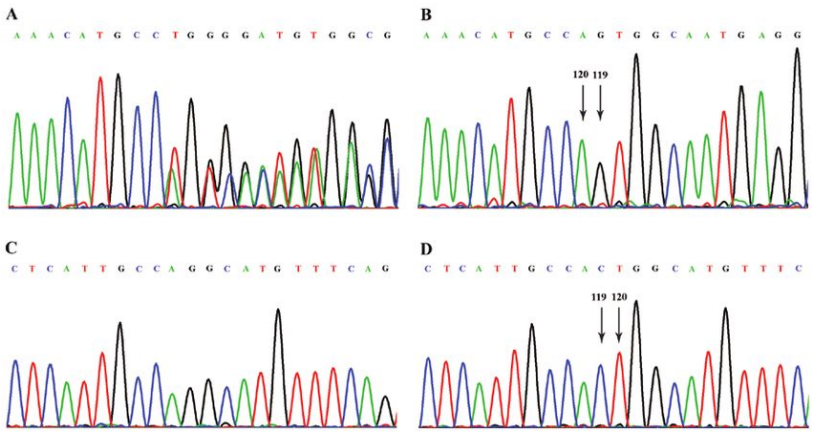
BME 105: Genetics in the Genomics Era

Lecture 4: Identifying genetic variants, part 1 (Sanger seq)



Instructor: Dr. Shelbi Russell,
Assistant Professor

Course website on canvas.ucsc.edu

Some slides modified from Angela Brooks

Lecture 4 Learning Objectives

- Become familiar with the types of DNA sequencing used for genome assembly (pre-2000):
- Be able to describe the steps in the polymerase chain reaction (PCR)
  - and how its used in genotyping
- Be able to articulate the similarities and differences between the chain-termination method and short-read sequencing by synthesis (i.e., Illumina sequencing)
- Understand why the chain-termination method (i.e., Sanger sequencing) is lower-throughput than short and long read technologies
  - Imagine the effort required to clone the entire human genome a few thousand bp at a time
- Be able to articulate what "shotgun" means in terms of genome sequencing
- Learn how genetic variants are categorized (SNP/SNVs, indels, CNVs, etc.)
- Be able to read a Sanger sequencing chromatogram and detect genetic variants
- Be able to describe how SNPs can be genotyped by microarray technology

How did we sequence the first (draft) human genome?

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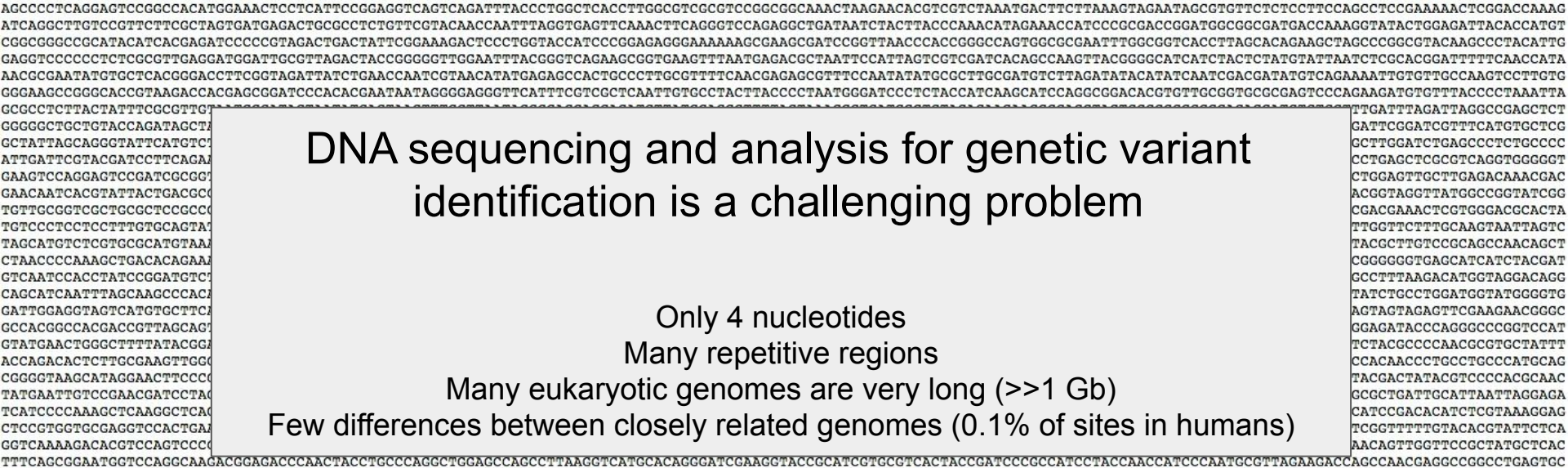
How did we sequence the first (draft) human genome?



DNA sequencing and analysis for genetic variant identification is a challenging problem

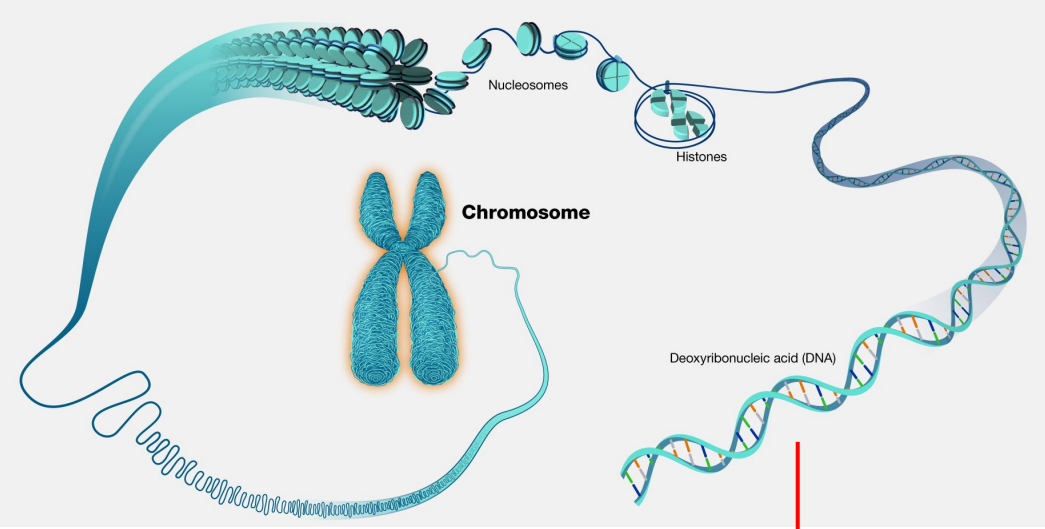
Only 4 nucleotides
Many repetitive regions
Many eukaryotic genomes are very long (>>1 Gb)
Few differences between closely related genomes (0.1% of sites in humans)

Roadmap: DNA sequencing and analysis for genetic variant identification

First-generation DNA sequencing (1977)

Second-generation DNA sequencing (~2000s)

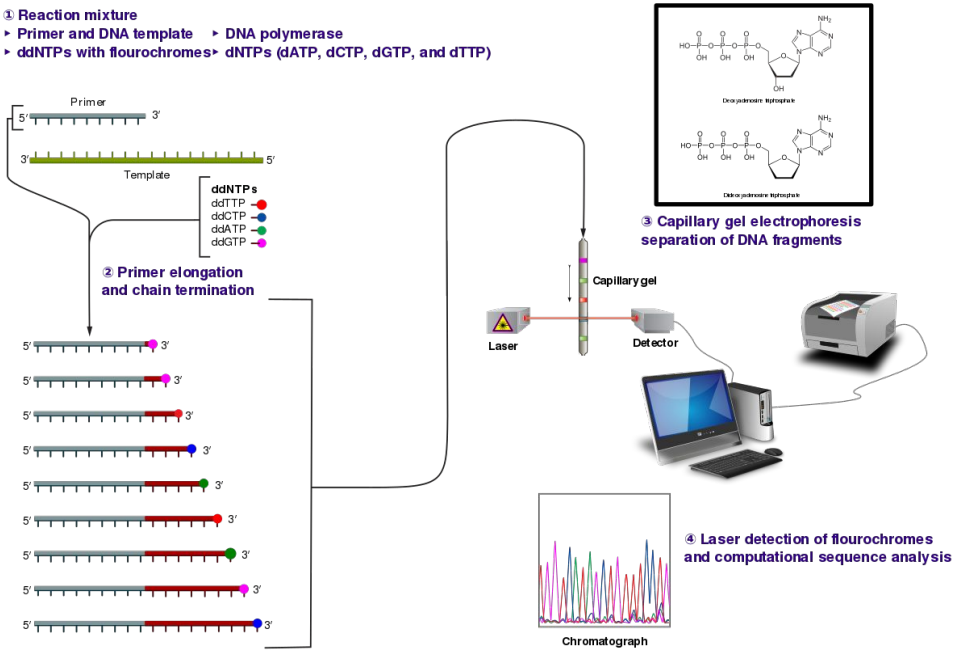
Third-generation DNA sequencing (~2010s)



Roadmap: DNA sequencing and analysis for genetic variant identification

First-generation DNA sequencing (1977)

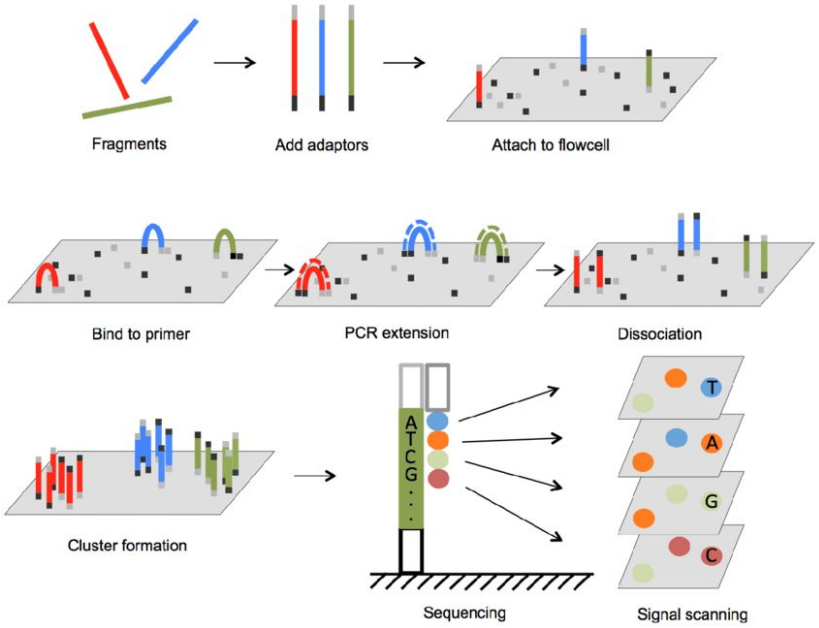
- Chain-termination method
- Sanger sequencing



Roadmap: DNA sequencing and analysis for genetic variant identification

First-generation DNA sequencing (1977)
- Chain-termination method
- Sanger sequencing

Second-generation DNA sequencing (~2000s)
- Massively parallel → high throughput
- One main platform until recently
  - Illumina sequencing (acquired Solexa)
  - [Roche 454 pyrosequencing - obsolete]
  - Illumina's main competitor: MGI



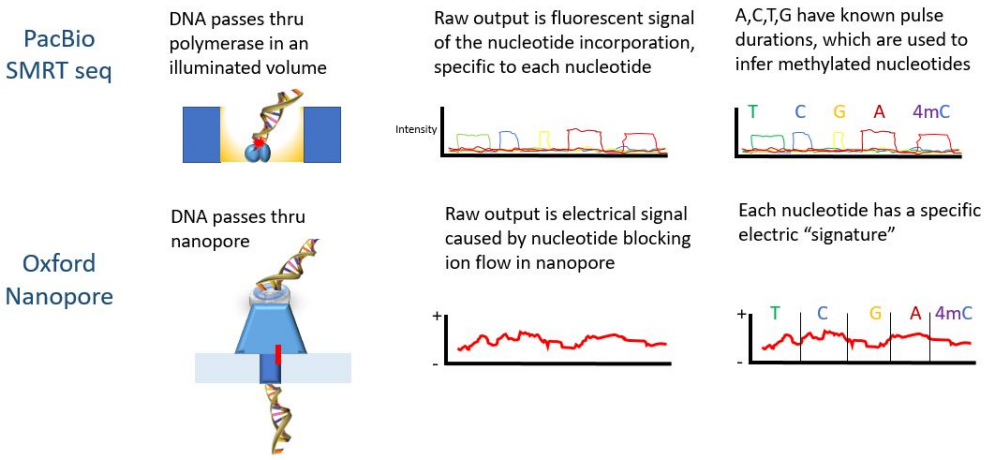
Illumina Sequencing

Roadmap: DNA sequencing and analysis for genetic variant identification

First-generation DNA sequencing (1977)
- Chain-termination method
- Sanger sequencing

Second-generation DNA sequencing (~2000s)
- Massively parallel → high throughput
- Illumina sequencing

Third-generation DNA sequencing (~2010s)
- Single-molecule
- Two main platforms:
  - Oxford Nanopore Technologies
  - PacBio

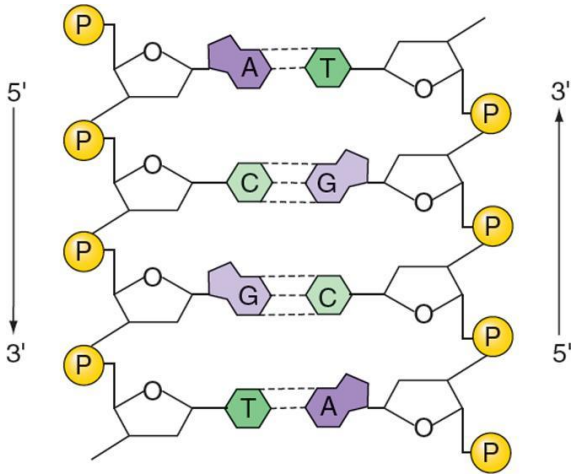


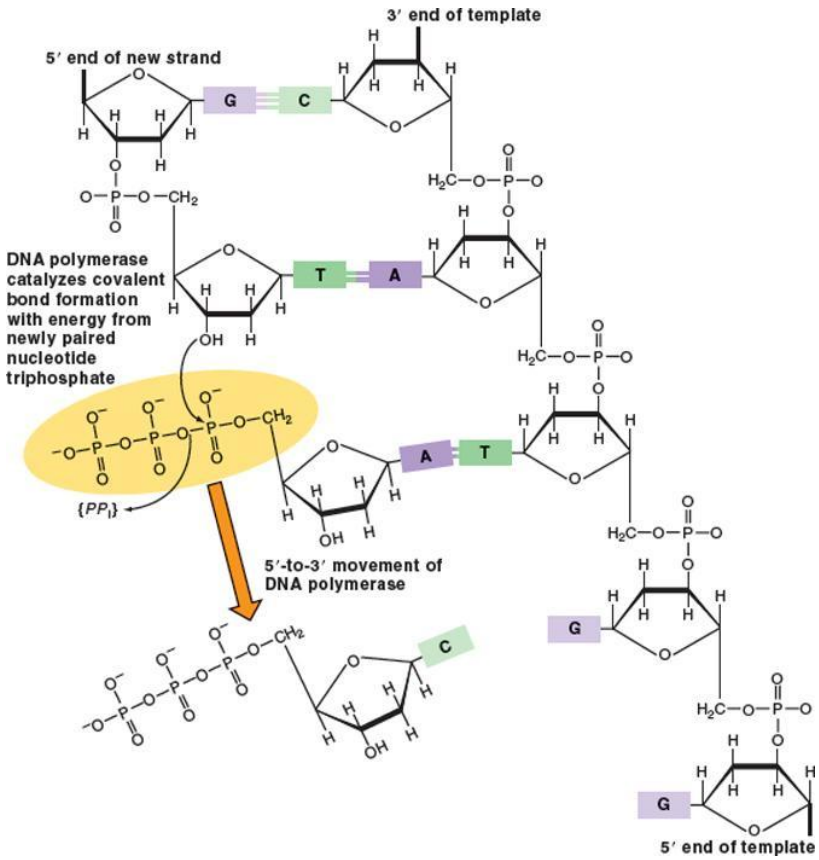
First-generation Sanger DNA sequencing

The chain-termination method

- Leverages DNA synthesis
- Requires PCR products as input

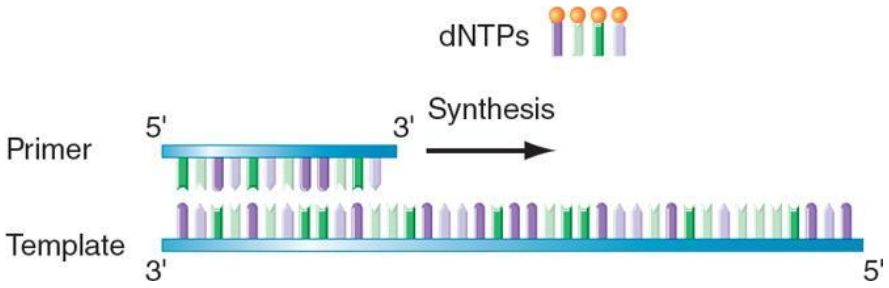
DNA base pairing

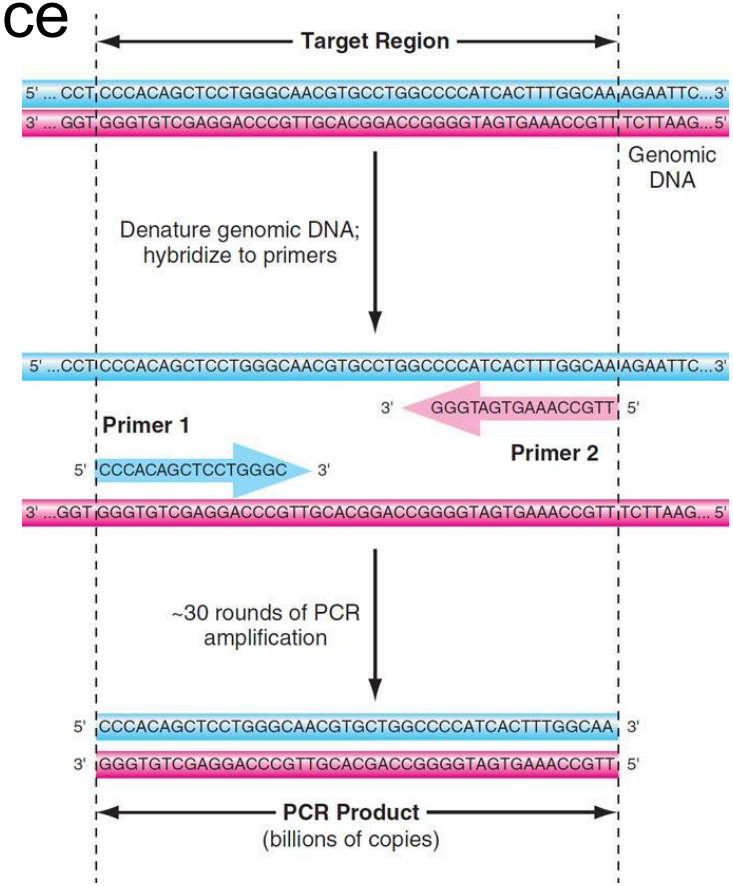




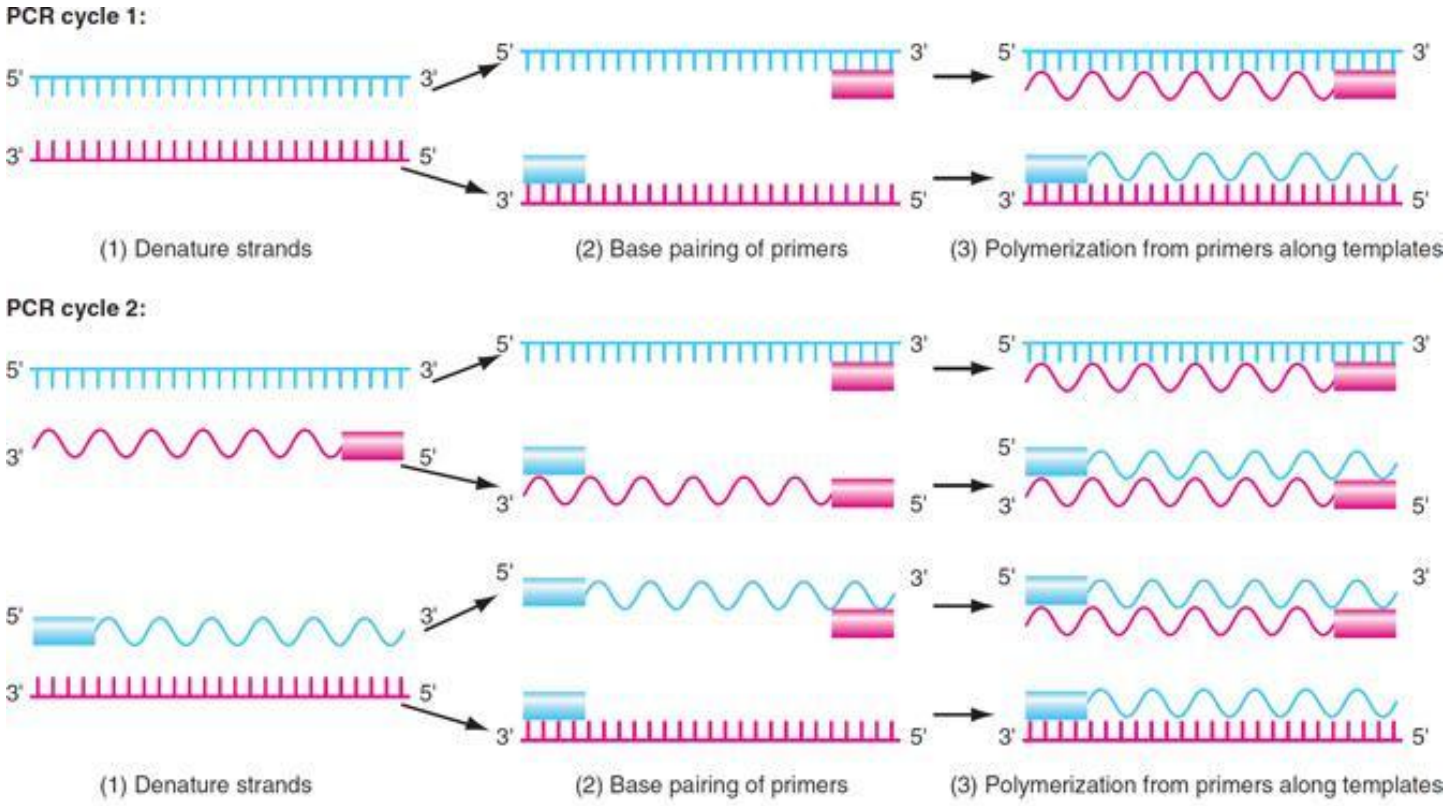
Polymerase chain reaction (PCR): Amplification of a target DNA sequence

PCR video: https://www.jove.com/science-education/10819/pcr

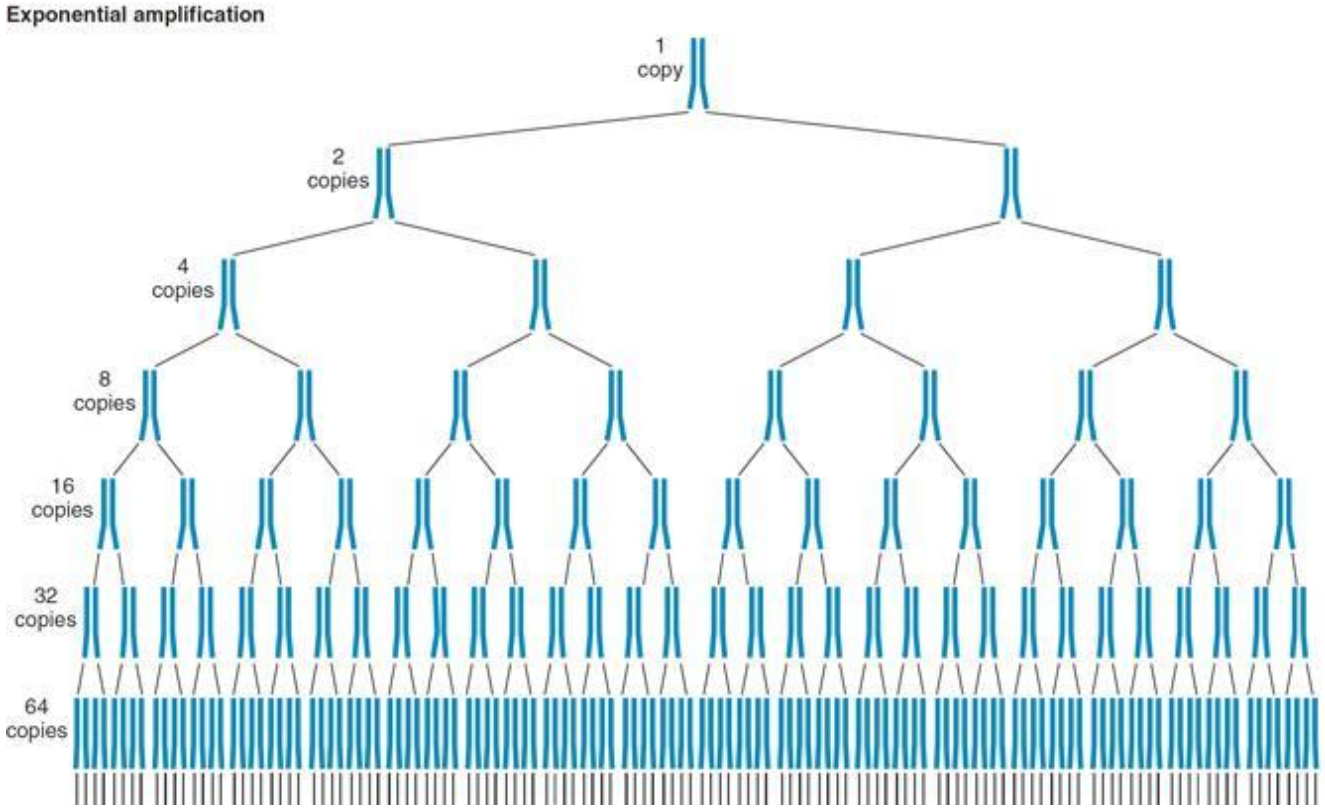




Polymerase chain reaction (PCR): Amplification of a target DNA sequence



Polymerase chain reaction (PCR): Exponential amplification of a target DNA sequence



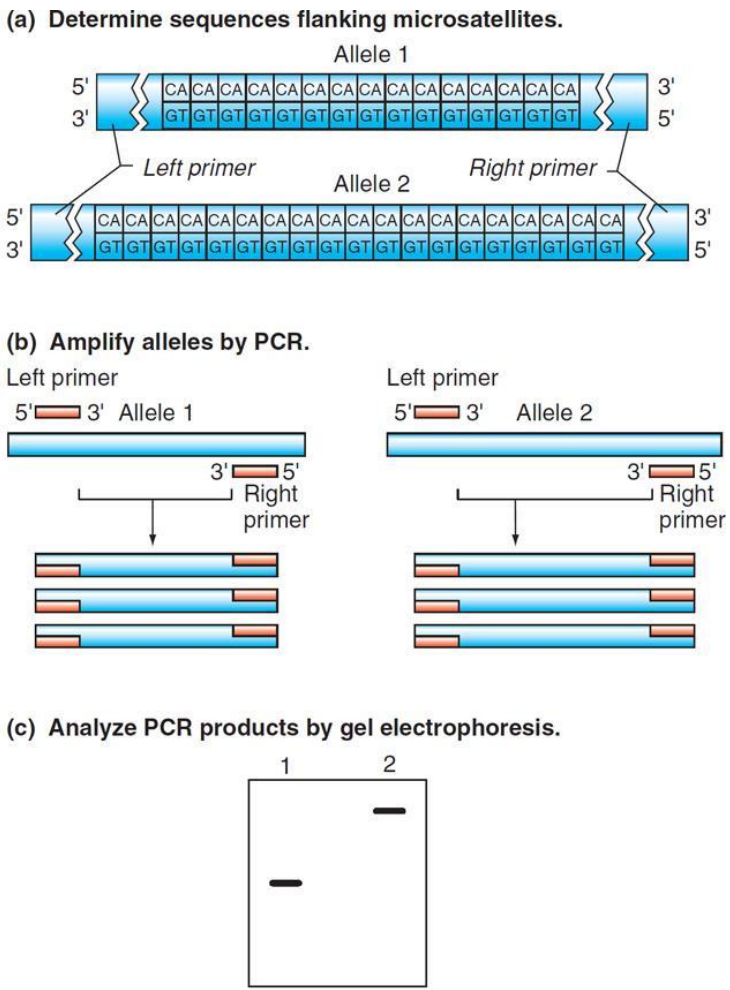
PCR alone can be used to detect allelic variation: microsatellites

SSR = single sequence repeat, also known as STRs = short tandem repeats

Microsatellites == SSRs

PCR primers in conserved regions, designed to straddle repeat polymorphism

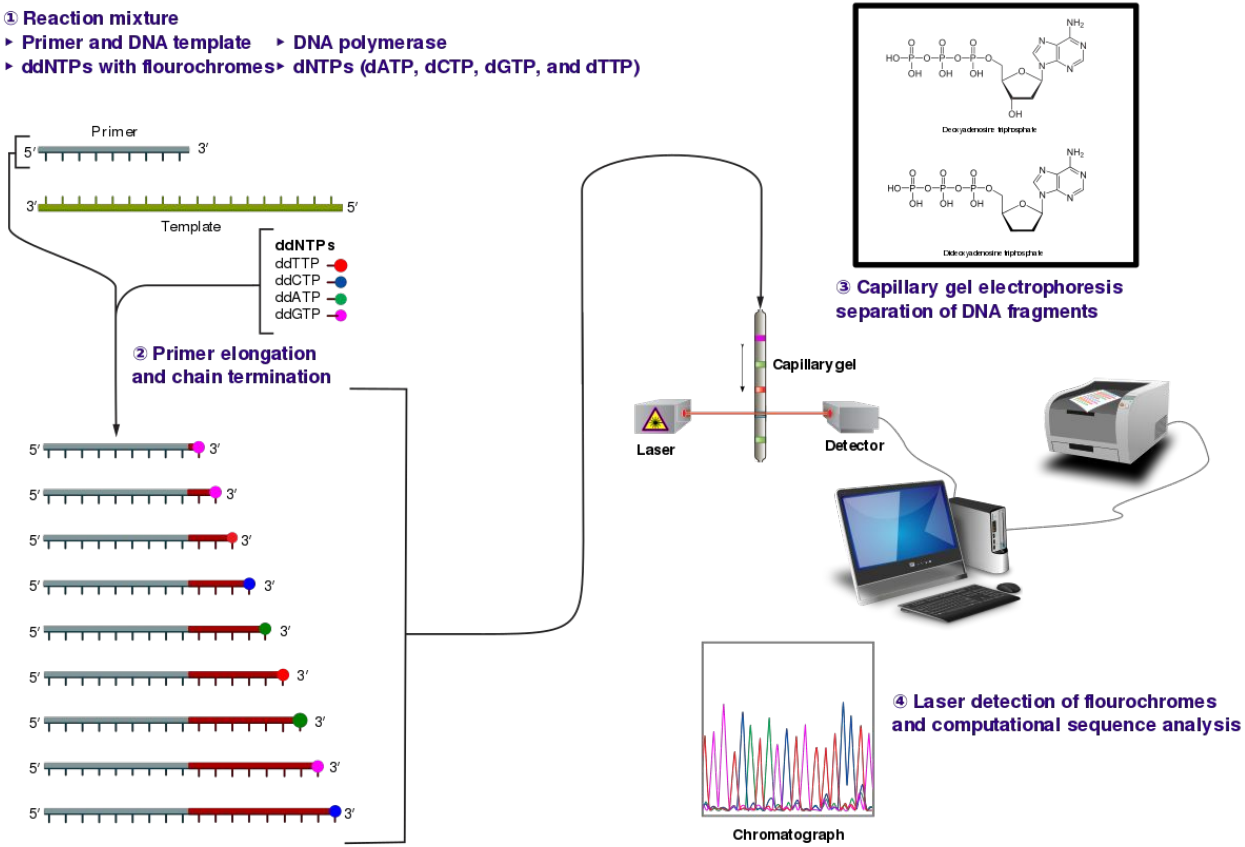
DNA amplicon length differences = repeat alleles



First-generation Sanger DNA sequencing

The chain-termination method

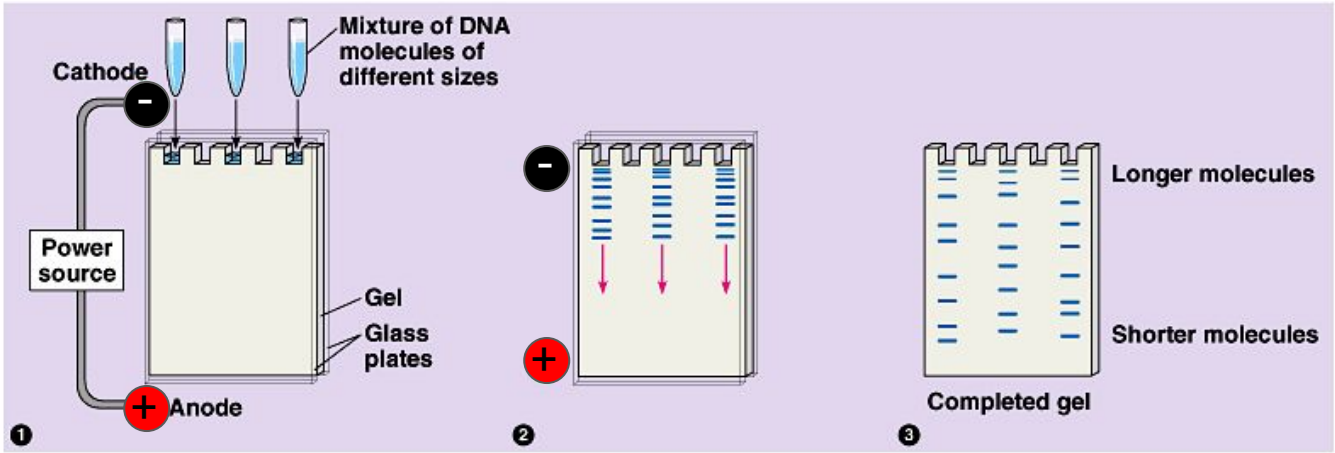
- Leverages DNA synthesis
- Incorporates fluorescent chain-terminator nucleotides to mark 3’ nucleotide
- Gel electrophoresis separates synthesis products by length
- Sequence is visualized by fragment order and color



Gel electrophoresis: Determine size of DNA molecules

- DNA is overall negatively charged due to the phosphate backbone
- Electric currents run negative to positive, or cathode to anode

→ You want to position the DNA with the cathode (-), so it runs to the anode (+)



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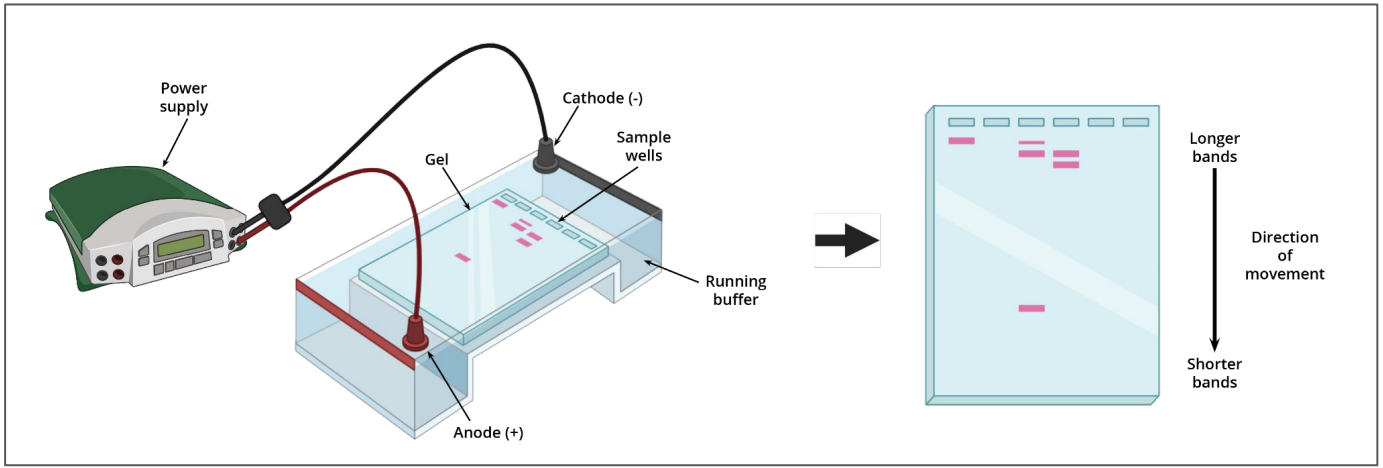


**On gel boxes, the cathode is black and the anode is red. You position the box black-to-red/top-to-bottom and load the DNA in the gel wells at the top (black end)

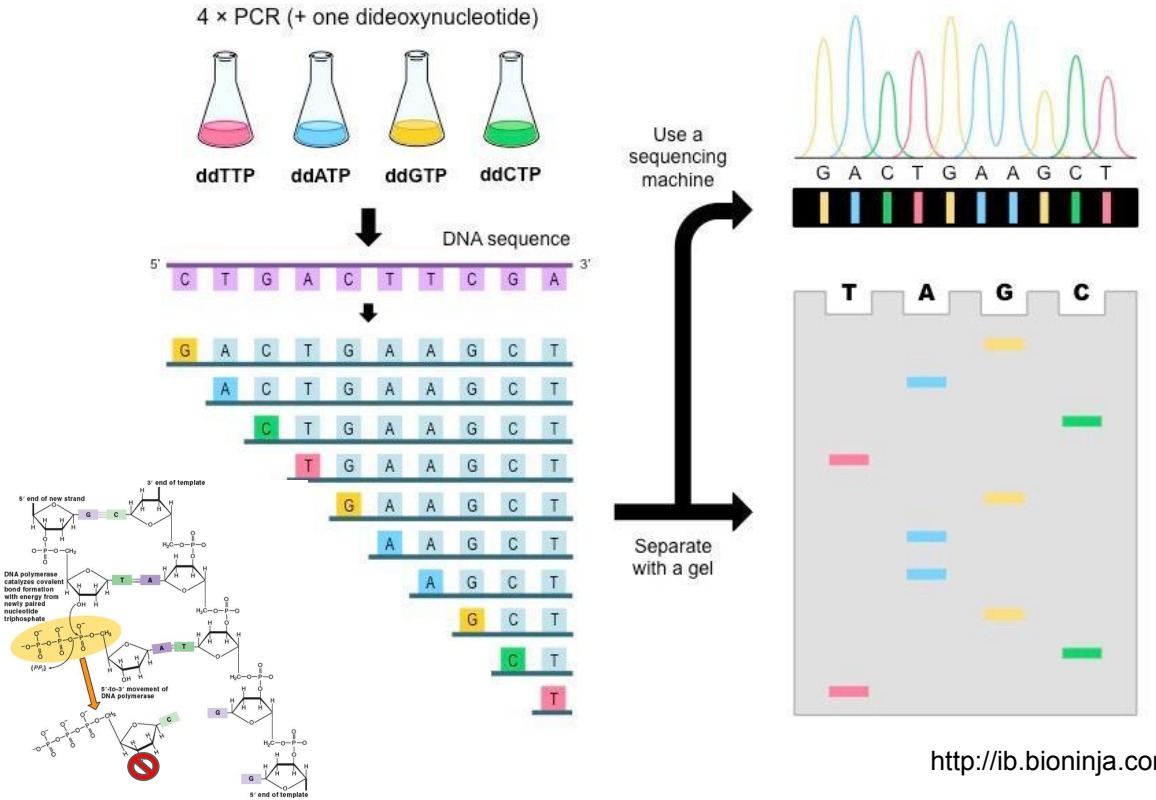
The DNA sequence is read in order by length and color

Sanger sequencing is similar to PCR, except you include a mix of dideoxynucleotides

- The sugar does not have an OH group, so the strand terminates

You obtain strands of different length

- The strand terminates at the included dideoxynucleotide sequence.



http://ib.bioninja.com.au

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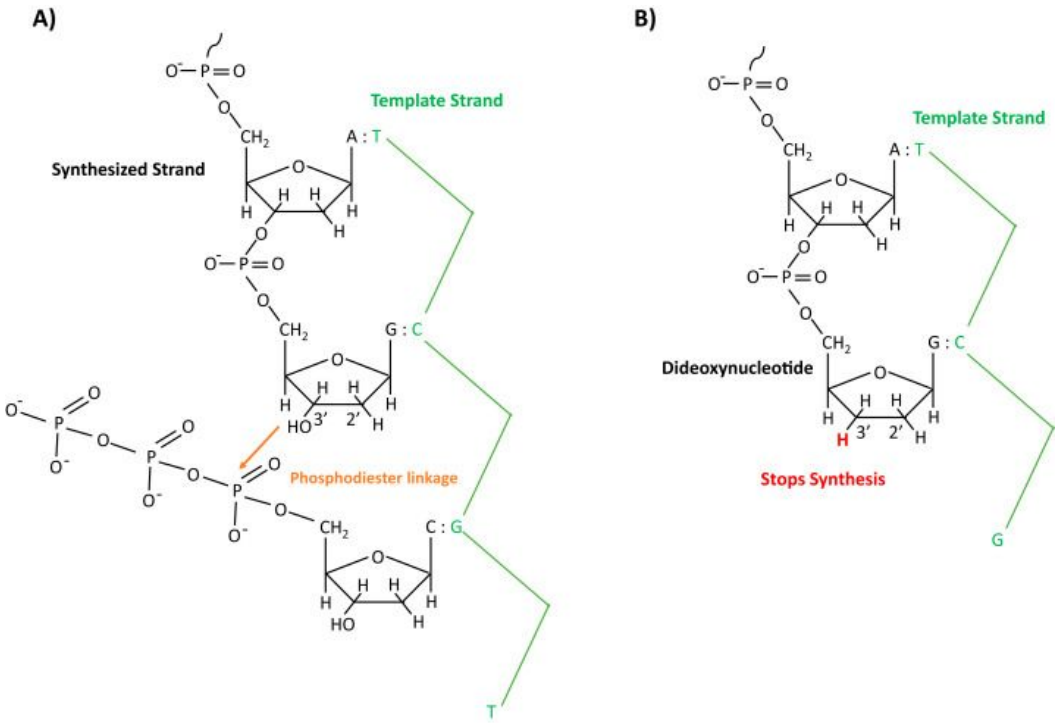
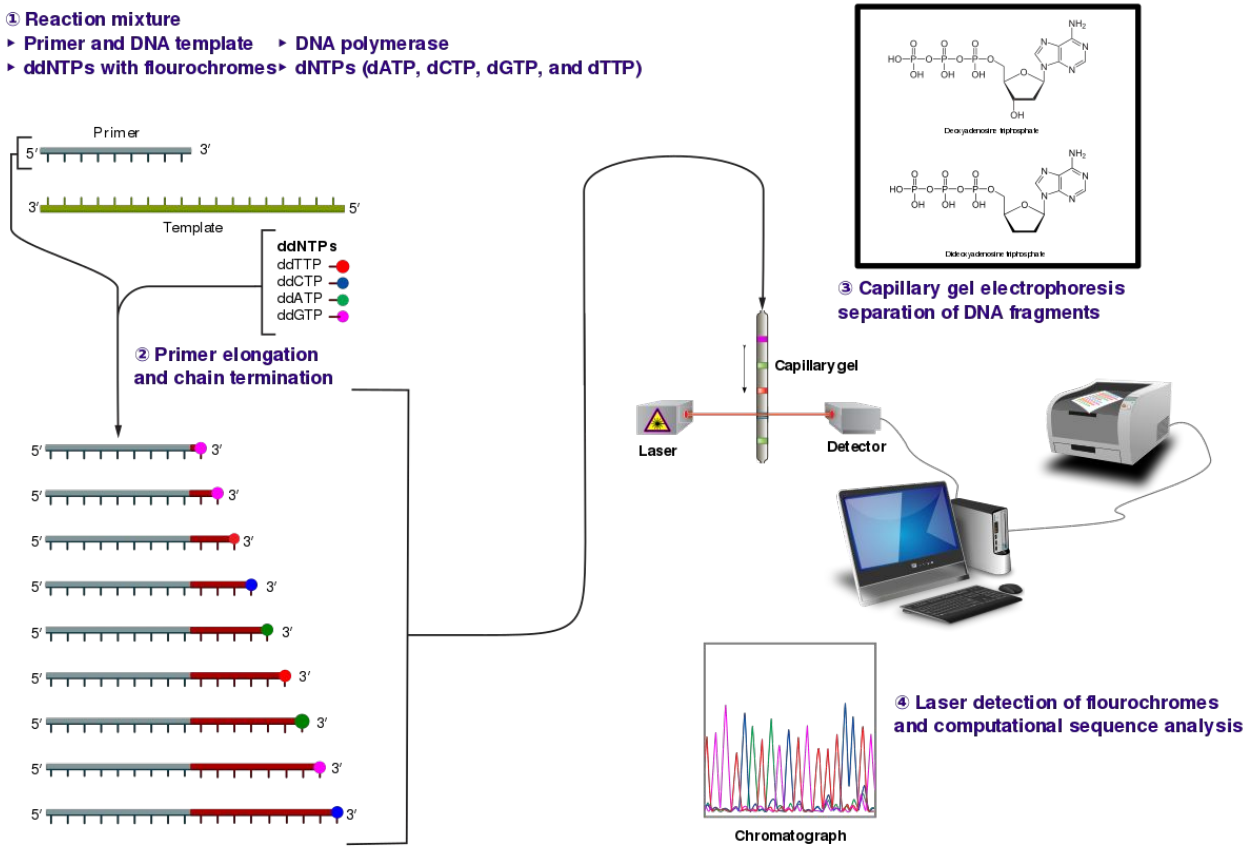


Shen 2023 Diagnostic Molecular Biology

First-generation Sanger DNA sequencing is still used

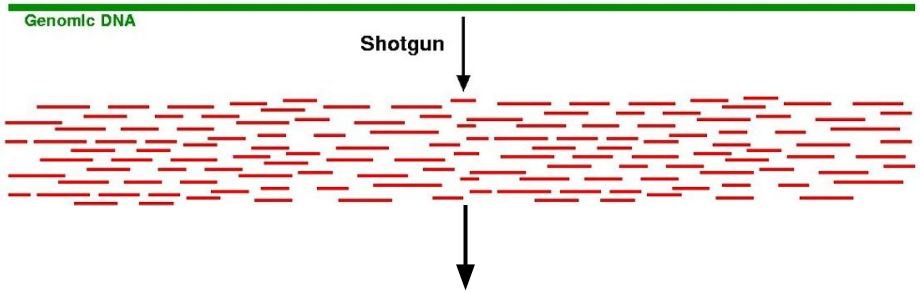
Used today to quickly (1-2 days):

- Confirm short-read sequencing results
- PCR product sequencing
- Plasmid sequencing



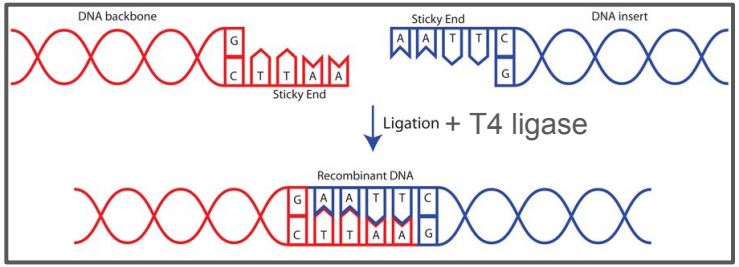
To assemble the human genome, a lot of cloning and computational biology was necessary

1. **Shotgun sequencing** → random fragmentation of the genome for sequencing
2. Ligate fragmented genomic DNA (gDNA) into plasmid/fosmid/cosmid inserts
   a. Provides primer sites for sequencing



Restriction sites

Sticky-end ligation mechanism



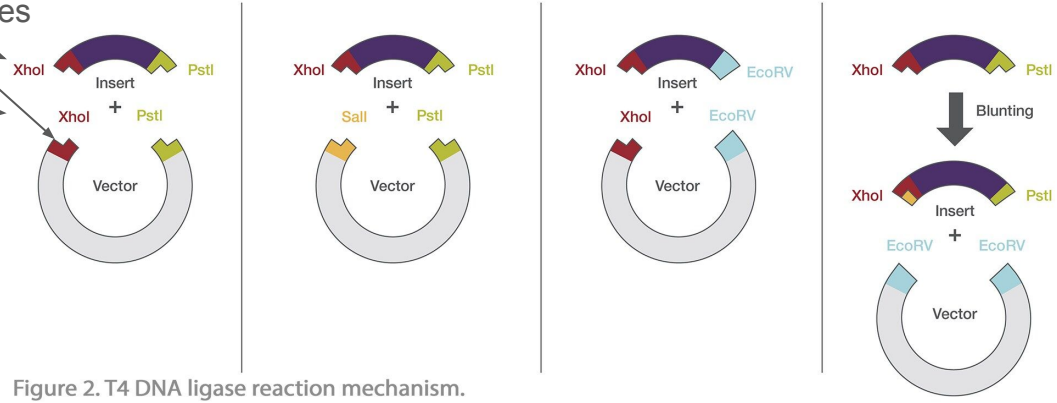
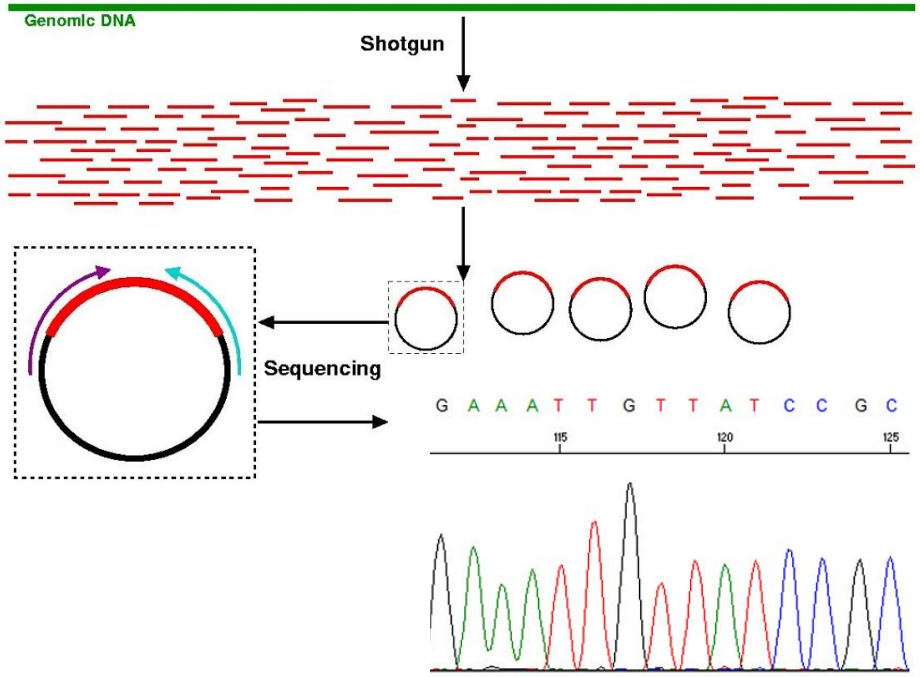


Figure 2. T4 DNA ligase reaction mechanism.

To assemble the human genome, a lot of cloning and computational biology was necessary

molecular cloning

1. **Shotgun sequencing** → random fragmentation of the genome for sequencing
2. Ligate fragmented genomic DNA (gDNA) into plasmid/fosmid/cosmid inserts
   a. Provides primer sites for sequencing
3. Grow ligated vectors in *E. coli* for purification and sequencing of pure traces
4. Repeat for the entire genome - must tile reads fairly deep across



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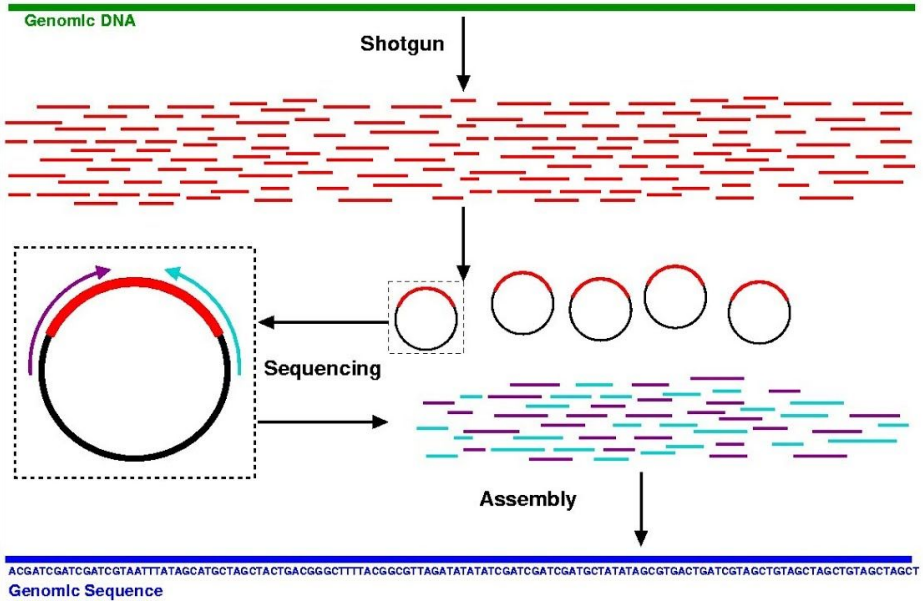
   a. Provides primer sites for sequencing
3. Grow ligated vectors in *E. coli* for purification and sequencing of pure traces
4. Repeat for the entire genome - must tile reads fairly deep across
5. Develop an assembly algorithm for overlapping sequencing reads and assembling contiguous tracts

*This was one of the major intellectual contributions from the Haussler lab at UCSC



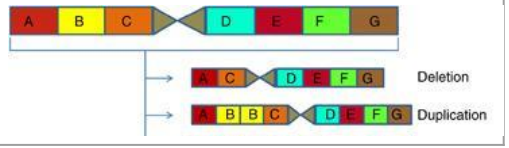
Categories of genetic variants

	Size	Frequency in human genome	Mechanism of generation	Example

SNP - Single nucleotide polymorphism	1bp	1 per 1kb	Mistakes in DNA replication; exposure to mutagens	`Allele 1 - CTGACTAGACGGC`
				`Allele 2 - CTGACAAGACGGC`
				`Allele 3 - CTGACCAGACGGC`

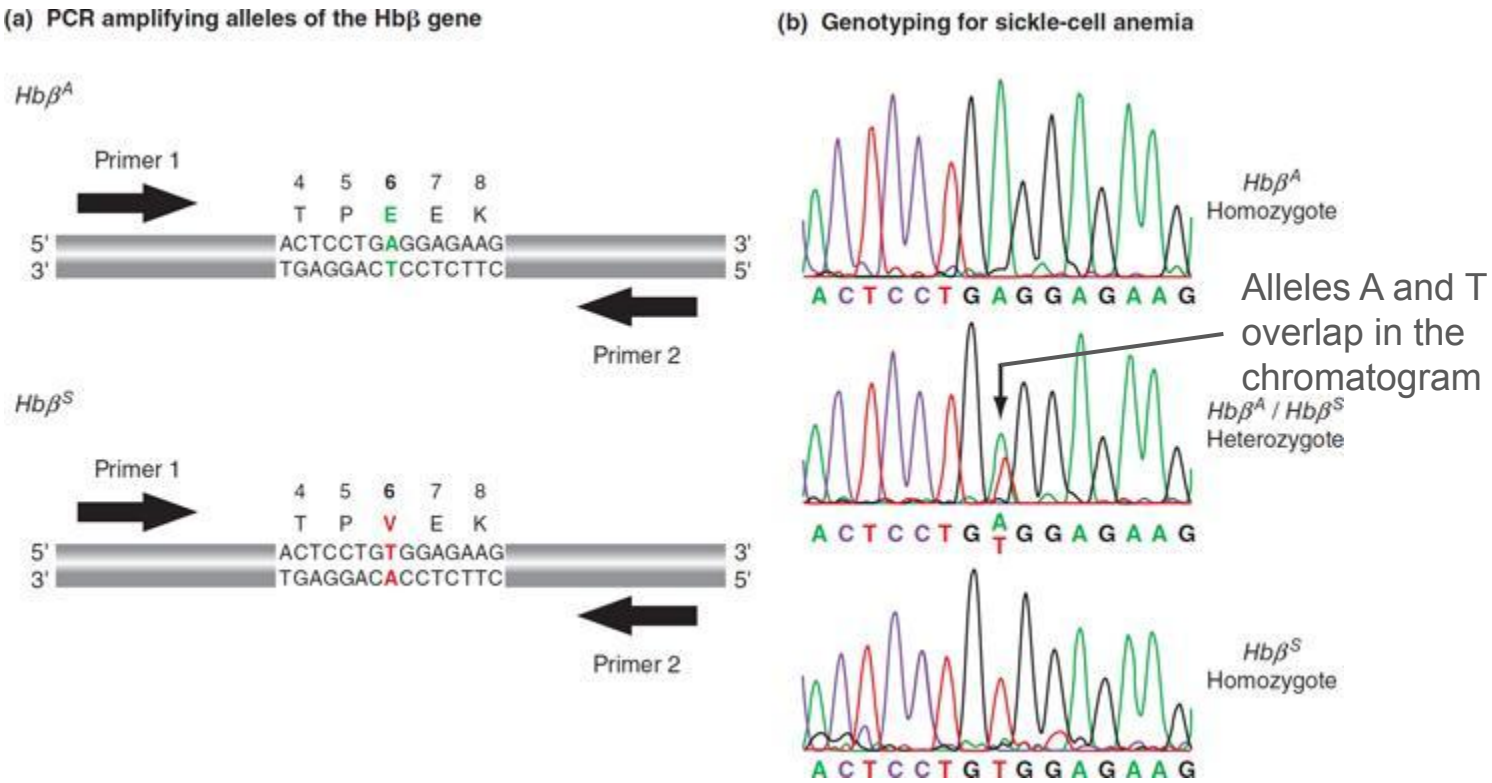
Indel - Insertion or deletion	~1-100bp	1 per 10kb	Mistakes in DNA replication (polymerase slippage - most common), recombination, and DNA damage repair	`Allele 1 - CTGACTAGACGGC`
				`Allele 2 - CTGAC-AGACGGC`
				`Allele 3 - CTGAC---ACGGC`

SSR - simple sequence repeat	1-10bp repeat unit	1 per 30kb	Once a repeat exists, easier to expand repeat due to slipped mispairing in DNA replication	`Allele 1 - CTGAGTGTGTGTCC`
				`Allele 2 - CTGAGTGTGTCCAG`
				`Allele 3 - CTGAGTGTCCAG`

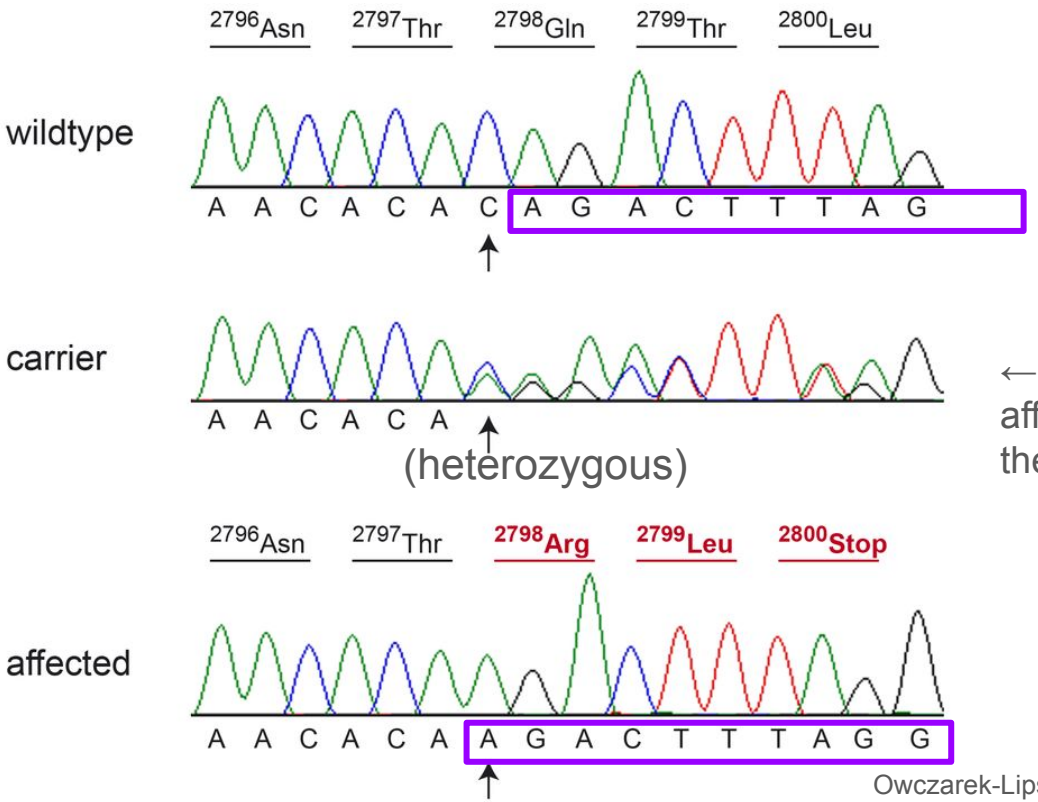
CNV - copy number variant	10bp-1Mb	1 per 3Mb	Unequal crossing-over in Meiosis I	

How are genetic variants identified from Sanger reads?

SNPs:



Single nucleotide deletion causes a stop codon



← deletion shifted the affected sequence to the left,
ruining chromatogram alignment

Owczarek-Lipska et al. 2013 PLoS One

Insertion-deletion (Indel) genotyping

- Indels change the sequence length
- In heterozygous samples, this shifts the alignment

Correct alignment

A

```
AAAGG-CAAGCAATAT
AAAGGGCAAGCAATAT
```

wrong alignment

B

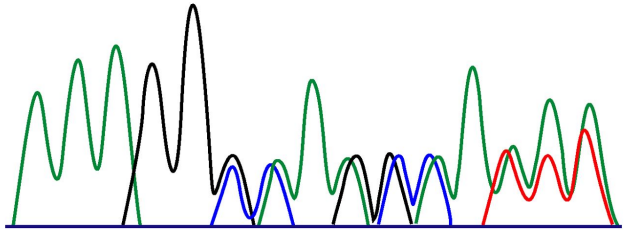
```
AAAGGCAAGCAATAT.
AAAGGGCAAGCAATAT
```

consensus

C

```
AAAGGSMARSMAWWW
```

D



deletion shifted the affected sequence

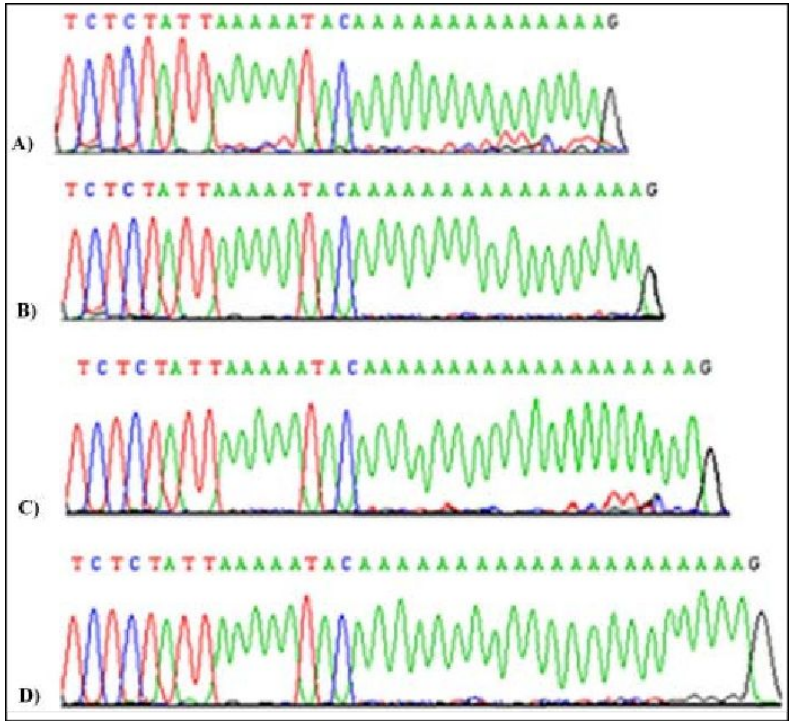
IUPAC symbols:

R = A or G
Y = C or T
K = G or T
M = A or C
S = C or G
W = A or T
B = not A
D = not C
H = not G
V = not T
N = any

Dmitriev and Rakitov 2008 PLoS Computational Biology

Simple sequence repeats (SSRs): microsatellite genotyping

- Also known as STRs (short tandem repeats)
- SSRs vary in length
- Consist of short repeating units
- Produced by slipped strand mispairing during DNA synthesis (aka replication slippage)

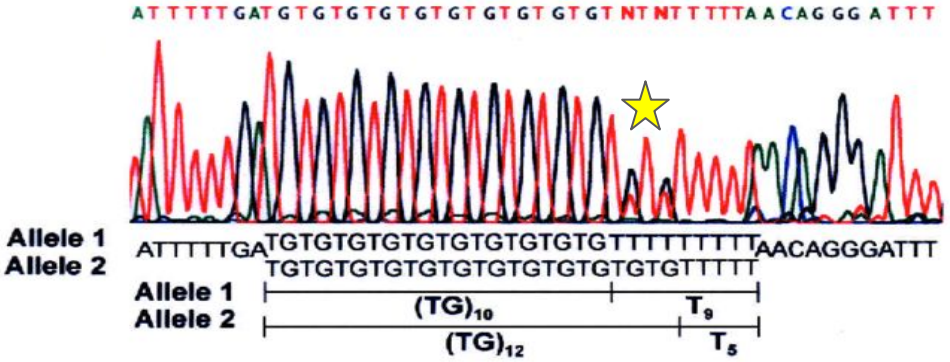


Colagar et al. 2016 Genetics in the 3rd Millennium

E.g., variation of poly(A) at the 3-end of the vitamin D receptor (VDR) gene

Simple sequence repeats (SSRs): microsatellite genotyping

Heterozygous SSR



Lucarelli and Grandoni 2002 BioTechniques

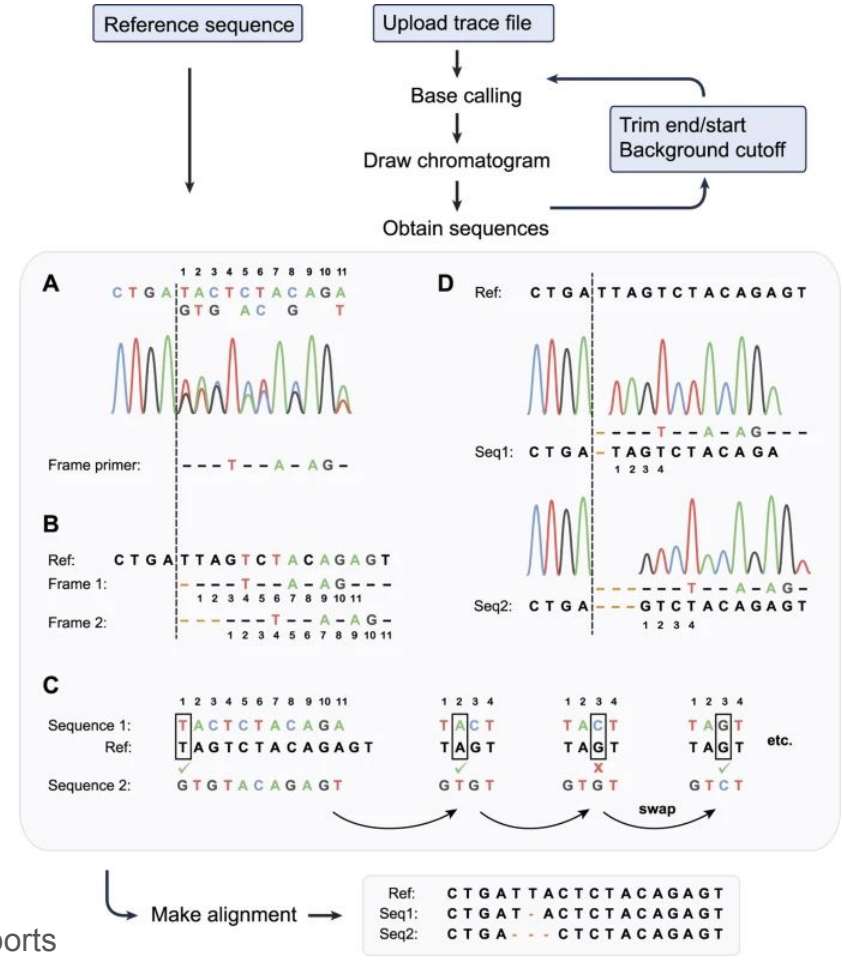
E.g., cystic fibrosis transmembrane conductance regulator (CFTR) gene (located at the intron 8/exon 9 splice acceptor site)

Computational chromatogram variant calling

While all of this is good to know how to do by eye, an algorithm makes the process more reproducible

This is only one example

You could even write your own, if you needed to!

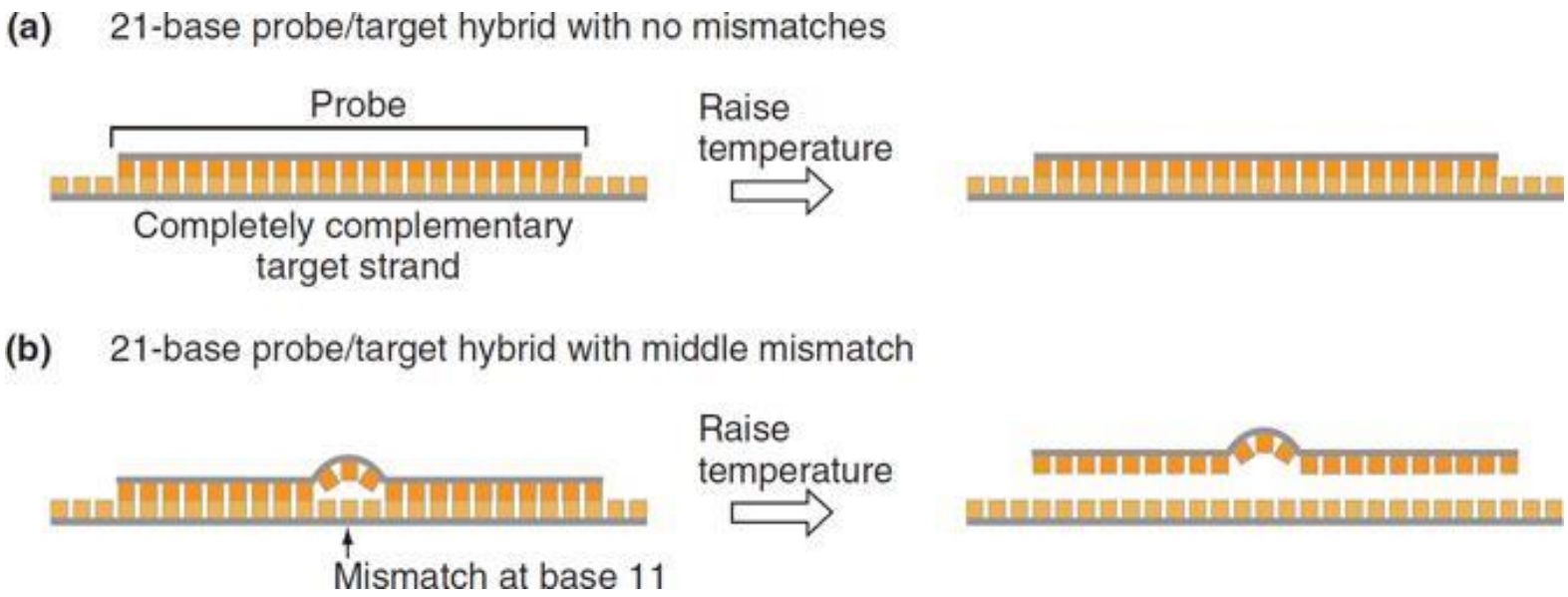


Dehairs et. al., 2016 Scientific Reports

Genetic variant detection by SNP microarrays

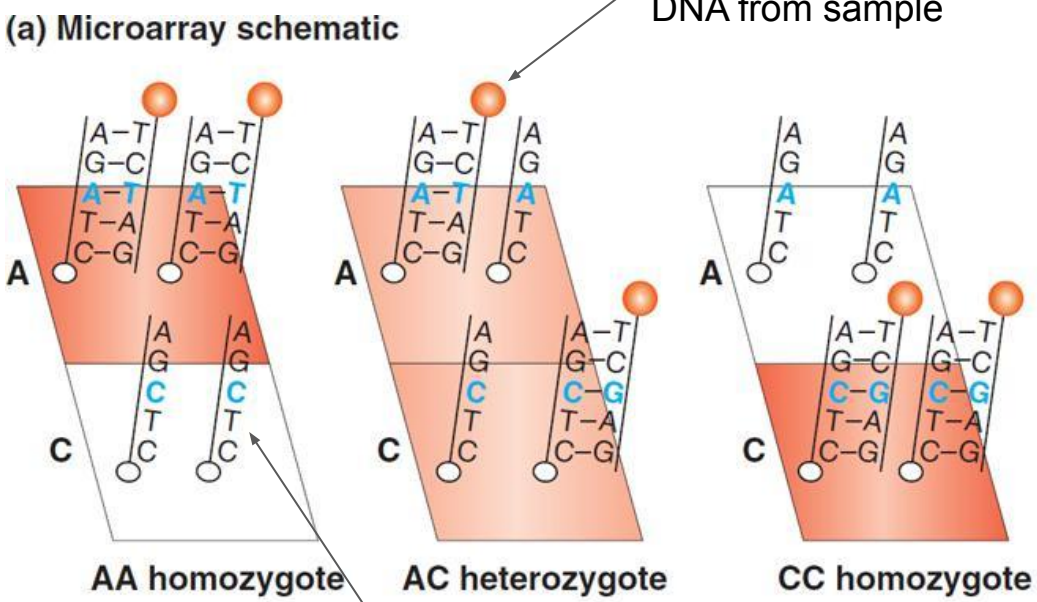
- Relatively cheap approach to genotype ~1 million SNPs, simultaneously
- Commonly used in large scale studies and cheaper commercial products (e.g. 23andMe, Ancestry)
- Consist of an array of DNA probes
  - Sequences complementary to genes/regions of interest
    - Disease markers
    - Markers correlated with ancestry
    - Other phenotypic traits

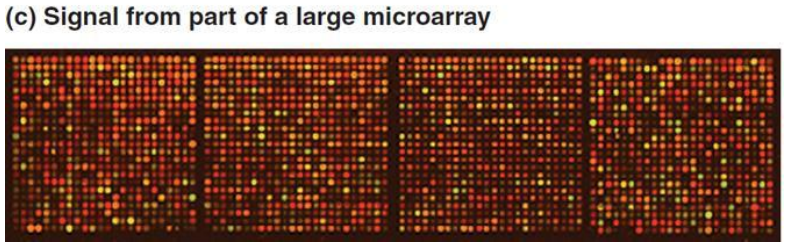
SNP microarray hybridization stringency



SNP microarrays

Fluorescently labeled
DNA from sample





**You have to know the sequence you want to print onto the microarray.

Pitfalls of Sanger sequencing and SNP microarrays

Must know primer/probe sequence

- Biased towards known sequences
- SNP arrays, sequence-specific primers, and vectors + primer sequences

Low-throughput

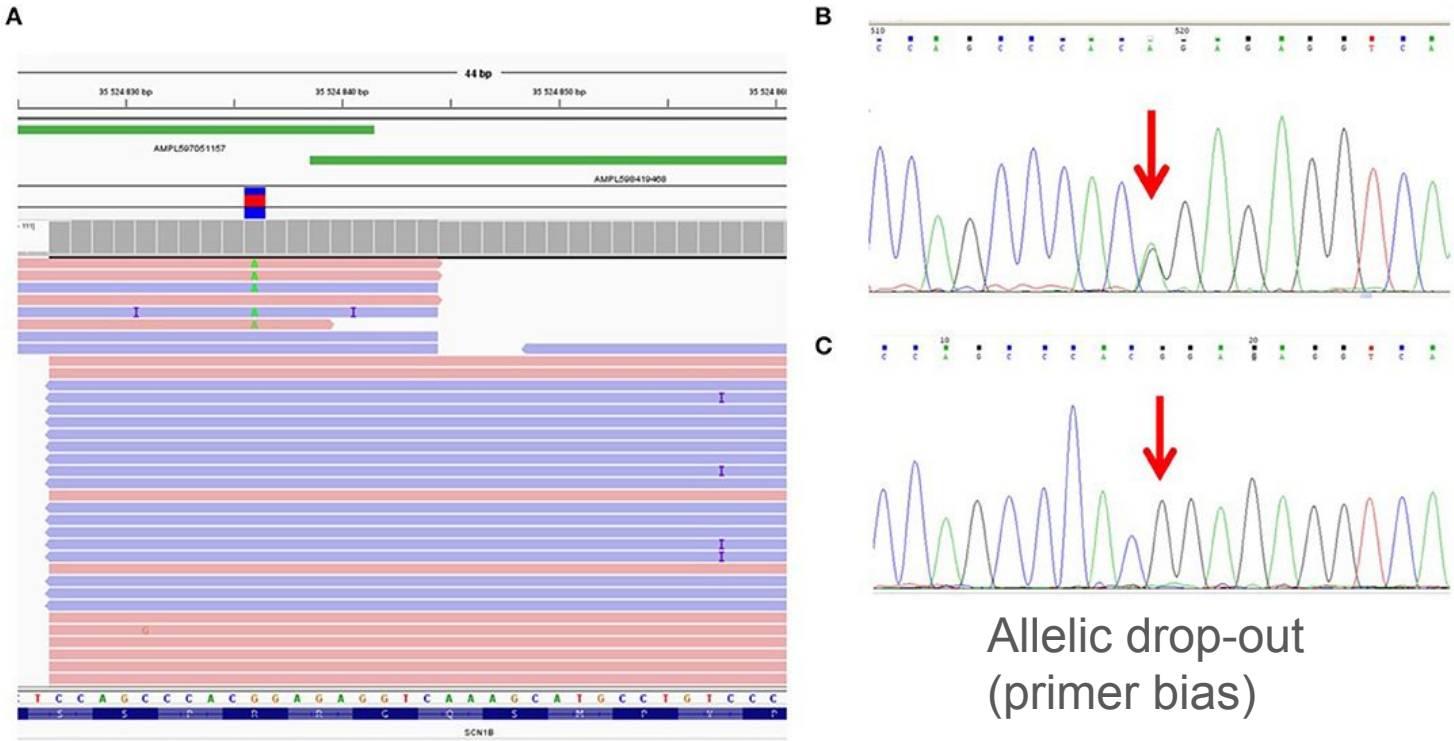
- Think of all the cloning required for unambiguous chromatograms
  - Across a complex genome (e.g., the human genome)
  - Across a diverse sample (e.g., the human microbiome)

- SNP chips only assay known variation (i.e., you must be able to design probes)

Chromatograms have limited resolution to disentangle mixed haplotypes

- Heterozygous samples
- Complex samples
- Contamination

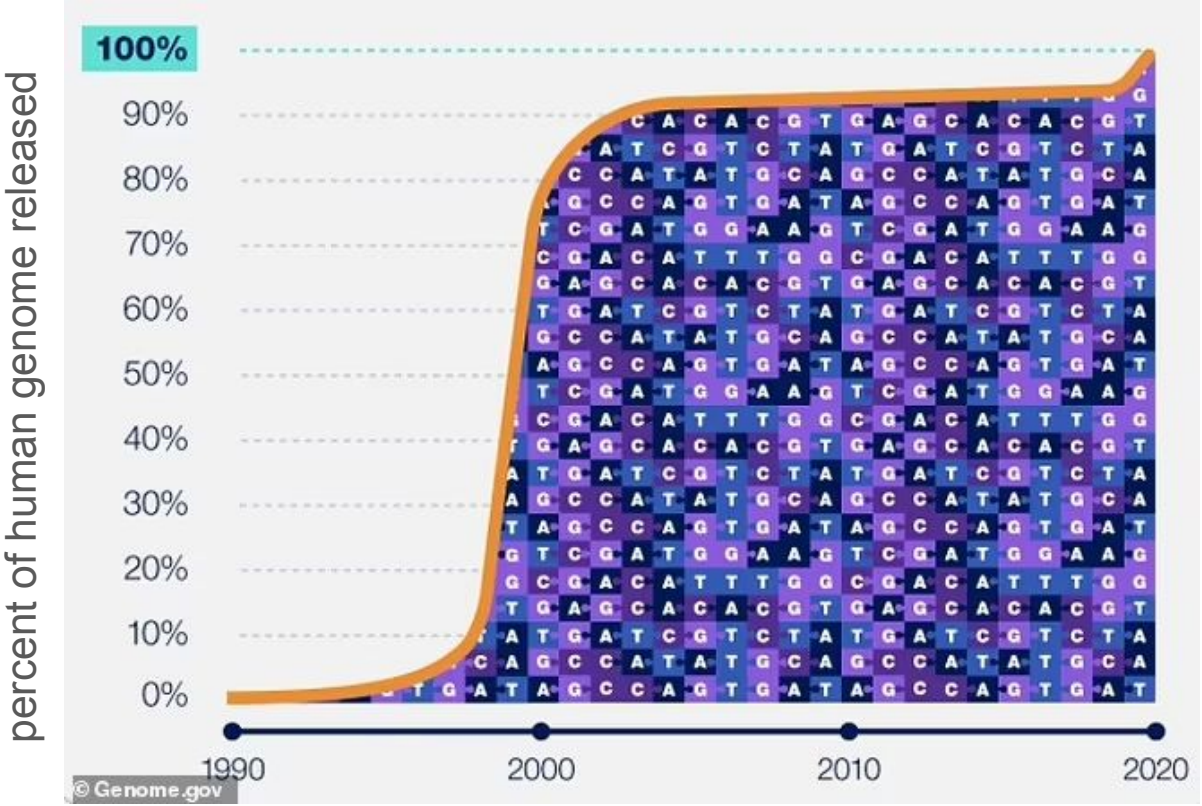
Next-generation sequencing approaches solve Sanger pitfalls



Allelic drop-out
(primer bias)

Shestak et al. 2021 Frontiers in Genetics

Next generation sequencing technologies enabled assembly of the last 10% of the genome



For Wednesday's class:

1. Continue reading in Genetics: From Genes to Genomes
   a. Finish reading Chapter 9.1-9.4,
   b. Start Chapter 11 (except 11.4),

2. Start working on Problem Set 4: Identifying genetic variation